Thermal Predictive Maintenance for Fast EV Charging Modules

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***Abstract*—Fast electric vehicle (EV) charging systems operate at high power densities, placing stringent thermal demands on power semiconductor devices. Silicon carbide (SiC) MOSFETs are widely adopted in fast chargers due to their superior electrical performance; however, their reliability is strongly constrained by junction temperature limits. Conventional thermal design approaches based on datasheet thermal resistances are insufficient to capture real operating conditions involving liquid-cooled cold plates and variable coolant flow rates.**

This paper presents a CFD-based thermal predictive framework for fast EV charging modules using ANSYS Icepak. A detailed steady-state conjugate heat transfer model of a three-device configuration of Infineon F3L8MR12W2M1HP B11 SiC MOSFETs mounted on a manufacturer-provided cold plate is developed. Junction temperature behavior is evaluated over a wide range of power dissipation and coolant flow conditions. The resulting thermal maps are used to determine minimum coolant flow requirements and to establish a predictive maintenance strategy capable of preventing thermal overstress through proactive power derating or controlled shutdown.

***Index Terms*—Fast EV charging, SiC MOSFET, thermal management, CFD, ANSYS Icepak**

I. Introduction

The rapid electrification of transportation has positioned electric vehicles (EVs) as a central solution to reducing greenhouse gas emissions and mitigating dependence on fossil fuels. As EV adoption accelerates worldwide, the deployment of reliable and high-power fast charging infrastructure has become a critical enabler of large-scale electrified mobility [1]–[3]. Fast charging stations must deliver large amounts of power within short time intervals while maintaining grid stability, operational efficiency, and user convenience. From a systems perspective, these stations act not only as high-power electrical loads but also as controllable assets interacting dynamically with the grid, renewable sources, and energy storage systems [1], [2]. Consequently, improving the performance, reliability, and sustainability of fast EV charging modules has emerged as a major research focus [3], [4].

At the heart of modern fast charging systems are high-efficiency power electronic converters capable of operating at elevated voltages, switching frequencies, and power densities. Silicon carbide (SiC) MOSFETs have become the preferred switching devices for next-generation EV chargers due to their superior electrical and thermal characteristics compared to conventional silicon-based IGBTs [4]–[7]. Their high breakdown voltage, fast switching capability, reduced switching losses, and elevated allowable junction temperatures enable converter efficiencies exceeding 98% and compact, high-power-density designs [4], [5]. These features make SiC devices particularly attractive for 800 V-class architectures and ultra-fast charging systems [8], [9]. However, the same high power density that enables compact designs also intensifies thermal stress within power modules, increasing heat flux and exacerbating thermal coupling among closely integrated chips [6], [8].

Thermal management has therefore become a defining constraint in the reliable operation of SiC-based EV charging systems. Elevated junction temperature is widely recognized as a critical factor influencing device reliability, lifetime, and failure mechanisms [9], [10]. In multi-chip power modules, high integration density and shared cooling paths introduce complex thermal interactions that cannot be adequately captured by simple lumped thermal resistance models [8], [11]. Advanced packaging and cooling strategies, including double-sided cooling, vapor chamber integration, and embedded liquid cooling architectures, have been proposed to mitigate excessive temperature rise and improve temperature uniformity [12]–[14]. These studies demonstrate significant reductions in junction temperature and thermal resistance, highlighting the importance of detailed electro-thermal analysis for high-power applications. Furthermore, transient thermal investigations of fast charging modules emphasize that increasing charging currents and power levels substantially aggravate heat generation, requiring carefully designed liquid cooling solutions to prevent overheating and long-term degradation [15].

Beyond conventional thermal design, there is growing interest in predictive and model-based thermal supervision techniques. Since direct measurement of junction temperature is impractical in most power modules, several estimation approaches have been developed based on electro-thermal coupled models, digital twins, neural networks, and data-driven methods [8]–[10]. These techniques aim to infer internal device temperatures from measurable electrical and thermal signals, enabling real-time condition monitoring and reliability assessment. In parallel, data-driven thermal modeling and anomaly detection frameworks have been explored for EV charging stations to enhance predictive maintenance and reduce operational risks [16], [17]. Such approaches shift thermal management from reactive protection mechanisms

toward proactive supervision strategies that anticipate thermal overstress before critical limits are exceeded.

Despite these advances, a gap remains between detailed physics-based thermal simulation at the device level and system-level operational decision-making in fast charging modules. Many existing works focus either on converter topology and performance comparison [4], [11] or on advanced cooling concepts [13], [14], while fewer studies translate simulation-derived thermal insights into practical predictive maintenance frameworks for multi-device assemblies. In high-power EV charging modules, multiple SiC MOSFETs mounted on a shared cold plate experience cumulative heat loading and coolant temperature rise along the flow path, making thermal coupling effects particularly significant. Accurately quantifying the relationship between power dissipation, coolant flow rate, and junction temperature distribution is therefore essential for defining safe operating envelopes and enabling proactive thermal control.

Motivated by these challenges, this work develops a CFD-based thermal predictive framework for a multi-device SiC MOSFET configuration representative of fast EV charging modules. By constructing detailed conjugate heat transfer models and performing parametric sweeps across power and coolant flow conditions, the study aims to bridge device-level thermal analysis with system-level predictive maintenance strategies. The resulting thermal maps provide quantitative guidance for coolant flow requirements, junction temperature supervision, and preventive control actions, contributing toward improved reliability and operational robustness in high-power EV charging infrastructure.

II. Device and System Description

The device analyzed in this study is the Infineon F3L8MR12W2M1HP B11 SiC MOSFET power module, commonly used in high-power EV charging and industrial power conversion applications [18]. The module consists of a SiC MOSFET die enclosed within a compact power package. Thermal management is provided by a liquid-cooled cold plate. For this study, three MOSFET modules mounted on the same cold plate were modeled to evaluate the thermal behavior of a multi-device configuration representative of practical fast charging systems, as shown in Figure 1.

III. Thermal Modelling Methodology

The thermal performance of the Infineon F3L8MR12W2M1HP B11 silicon carbide (SiC) MOSFET was evaluated using steady-state conjugate heat transfer simulations in ANSYS Icepak (ANSYS Electronics Desktop). The objective of the modeling approach was to establish a quantitative relationship between total dissipated power, coolant flow rate, and junction temperature, enabling prediction of cooling requirements necessary to maintain each device below its maximum allowable junction temperature of 125 °C [18]. In this study, three identical MOSFETs were modeled on a common cold plate to capture cumulative heat loading and thermal interaction effects.

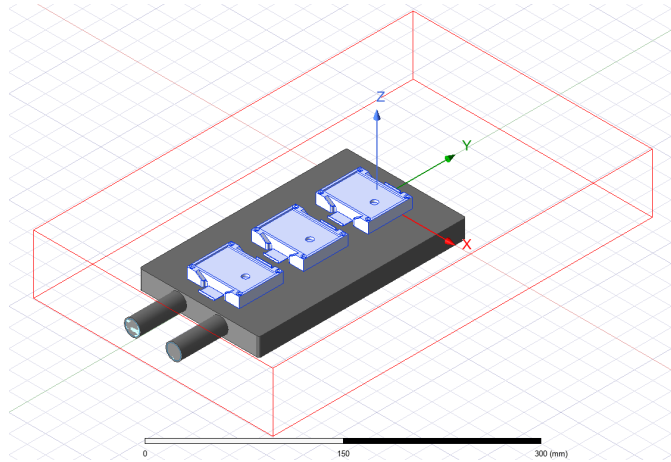

Fig. 1. Overall CFD model showing the three-MOSFET assembly on the cold plate and the surrounding air region used for boundary condition assignment.

A. Geometry Preparation and Simplification

The MOSFET package geometry and cold plate geometry were obtained directly from official design resources provided by their respective manufacturers. The MOSFET package model was downloaded from Infineon's design resources for the F3L8MR12W2M1HP_B11 device [19], while the baseline cold plate geometry was obtained from the manufacturer resources for the Wieland Microcool CPE-4000 series cold plate [20]. Both geometries were imported into Onshape for preprocessing and simplification prior to CFD analysis.

Geometric simplifications were applied to reduce mesh complexity while preserving dominant heat transfer paths. These included removal of non-functional fillets, internal and external holes not involved in heat transfer, and minor surface details that do not influence conduction or convection. MOSFET pins were removed, as electrical functionality was not modeled, and the baseplate geometry was simplified as shown in Fig: 2.

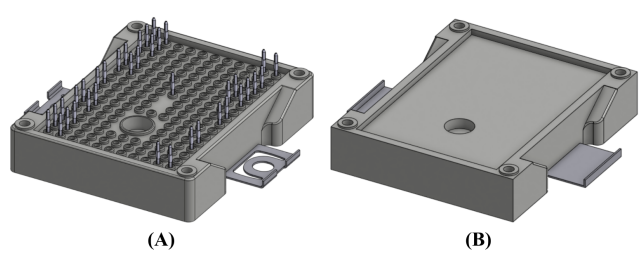

Fig. 2. MOSFET package geometry: (A) original manufacturer-provided model from Infineon, and (B) simplified model used for CFD.

The manufacturer-provided cold plate geometry represented the solid plate with internal flow channels defined as cavities rather than an explicit solid fluid domain. Since ANSYS Icepak requires a closed solid volume to define the coolant region, the fluid domain was first reconstructed using Boolean operations. The original geometry included two parallel straight channels with inlet and outlet ports on the

same face; a connecting channel was added at the distal end to form two U-shaped flow paths, accurately representing the physical coolant route. Finally, the entire cold plate geometry was simplified to reduce mesh complexity while preserving the main flow paths, as shown in Fig: 3. These steps ensured realistic velocity and temperature distributions under the combined heat load of the three MOSFETs.

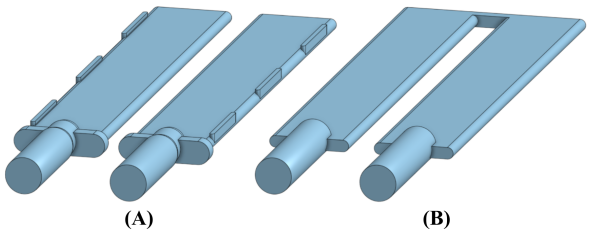

Fig. 3. Coolant flow paths in the cold plate: (A) reconstructed solid domain using the manufacturer-provided model, and (B) solid domain after connecting channels and geometric simplifications.

The simplified MOSFET package, cold plate geometry, and reconstructed coolant solid domain were then imported into ANSYS Icepak. Within the simulation environment, the geometries were assembled and organized appropriately. The MOSFET model was duplicated along the mounting direction to create the three-device configuration on the common cold plate, after completing material and boundary assignments for the initial device, as shown in Figure 1.

B. MOSFET Package Decomposition and Material Assignment

The imported MOSFET geometry represented the external plastic encapsulation as a solid body. Additional internal regions were explicitly created within ANSYS Icepak to represent the baseplate, inner mold compound, and the SiC die, as shown in Fig. 4, allowing accurate modeling of conduction paths from the junction to the cold plate.

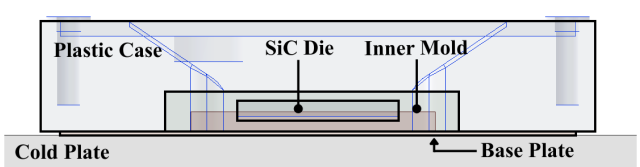

Fig. 4. Section view of the MOSFET package showing the internal structure.

Material properties were assigned based on commonly accepted approximations for each component. The SiC die was modeled using aluminum nitride (AlN) as a thermally equivalent material. All assigned materials and thermophysical properties are summarized in Table I.

TABLE I
ASSIGNED MATERIALS PROPERTIES

Component	Material	k (W/m·K)	ρ (kg/m^3)	c_p (J/kg·K)
MOSFET case	Plastic	0.23	1200	1050
Base plate	Aluminum (extruded)	205	2800	900
Inner mold	Mold compound	0.8	1900	795
SiC die	AlN (equivalent)	120	3250	734.3
Cold plate	Aluminum 1100	218	2710	900
Coolant	Water @ 280 K	0.582	1000	4198
Air Region	Air (default)	0.0261	1.161	1005

C. Heat Source Definition and Contact Modeling

Heat generation was applied exclusively to the SiC die region using a total power thermal condition. The applied power was defined as a parametric variable ("Power") to enable automated sweeps across multiple operating conditions. All other package components were modeled as passive thermal solids.

A thermal contact resistance of 0.067 K/W was assigned at the interface between the MOSFET baseplate and the cold plate to represent realistic mounting conditions with thermal interface material applied. This value was selected based on prior experimental validation work and reflects typical performance of commercial thermal pastes under moderate clamping pressure.

Temperature monitors were placed on multiple components, with primary focus on the maximum temperature of the SiC die, which was used to define the junction temperature.

D. Boundary Conditions

The air region surrounding the assembly was automatically generated by ANSYS Icepak upon model import. The default padding configuration was retained in all directions, except along the axis intersecting the coolant inlet and outlet, where the padding offset was reduced from the default 50% to 0% to prevent unnecessary domain extension. Five faces of the air region were assigned pressure-opening boundary conditions at ambient pressure, allowing bidirectional flow and natural convection effects. Gravity was enabled to capture buoyancy-driven airflow.

The coolant inlet velocity was defined using a parametric variable ("FlowRate") with units of m/s, while the outlet was assigned an ambient pressure boundary. The coolant channel diameter was 15.6 mm, and inlet velocities were later converted to volumetric flow rates for analysis.

E. Mesh and Solver Configuration

The computational mesh resolution was selected at an intermediate level between coarse and fine presets to balance accuracy and computational cost while remaining within student-license mesh constraints.

Steady-state simulations were performed with a maximum of 100 solver iterations. The flow regime was modeled as turbulent to capture enhanced convective transport at higher flow rates. Radiation effects were neglected due to the dominance of forced convection under liquid-cooled conditions, and gravity

was included. The solution type was configured to solve for coupled temperature and flow fields.

F. Parametric Study Definition

A parametric sweep was implemented using ANSYS Optimetrics to evaluate the combined effects of power dissipation and coolant flow. The power dissipation was varied linearly from 50 W to 600 W in 50 W increments (12 levels), while coolant inlet velocity was swept across 18 discrete values ranging from 0 m/s (natural convection) to 5 m/s. This resulted in a total of 216 steady-state operating points.

The combined parametric sweep covered natural convection through high-flow forced convection conditions, enabling construction of a flow–power operating envelope. This dataset forms the basis for the predictive maintenance methodology presented in subsequent sections.

IV. Results and Analysis

A. Representative Temperature Distribution

To illustrate the spatial thermal behavior of the three-device configuration, Figure 5 presents the steady-state temperature distribution for a representative operating point corresponding to 200 W heat dissipation per MOSFET and a coolant inlet velocity of 0.018 m/s (total electrical input power $\approx$ 30 kW).

The 3D surface temperature distribution in Fig. 5(a) illustrates localized hot spots near the die region and relatively uniform heat spreading within the cold plate. The cross-sectional view in Fig. 5(b) further highlights vertical heat conduction from the SiC die into the cold plate. The three devices exhibit thermal coupling due to shared conduction paths and coolant temperature increase.

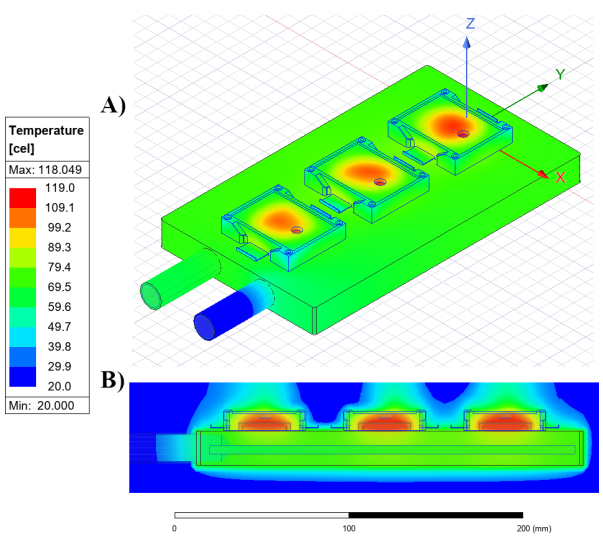

Fig. 5. Thermal contour plots of the three-device assembly at a representative high-power operating point (200 W dissipation per module, 0.21 L/min coolant flow): (A) Isometric surface temperature distribution, and (B) Cross-sectional temperature profile through the SiC MOSFET center-line.

B. Junction Temperature Versus Power

The steady-state simulation results show an increase in the maximum junction temperature with increasing power dissipation across all investigated coolant flow rates. Figure 6 presents the variation of junction temperature of a single MOSFET within the three-device configuration as a function of dissipated power for representative coolant flow velocities.

For a fixed coolant flow rate, the junction temperature exhibits an approximately linear dependence on power dissipation. Increasing the coolant flow rate reduces the slope of the temperature–power relationship, reflecting improved heat removal capability.

At lower coolant flow velocities, the junction temperature approaches the allowable limit at comparatively lower power dissipation levels, whereas higher flow velocities enable sustained operation at elevated power. These results demonstrate the coupled influence of power dissipation and coolant flow on junction temperature and motivate the use of flow-dependent thermal operating limits.

Although only one representative MOSFET temperature is shown for clarity, the temperature difference between devices was evaluated across all operating points. For most conditions, the temperature spread between MOSFETs remained modest (median $\approx$ 2–3 °C), indicating relatively uniform cooling. However, under certain high-power and low-flow conditions, localized temperature differences exceeding 100 °C were observed, highlighting the importance of flow-dependent operating limits in multi-device configurations.

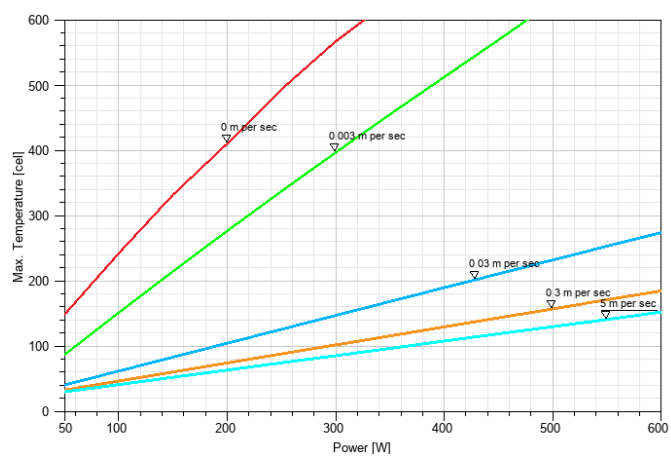

Fig. 6. Maximum junction temperature of a single SiC MOSFET on a three-MOSFET configuration as a function of dissipated power for representative coolant flow velocities.

C. Coolant Flow Requirements

Figure 7 illustrates the maximum junction temperature of a single MOSFET in the three-device configuration as a function of coolant flow rate for selected power dissipation levels. For each power level, increasing the coolant flow initially results in a significant reduction in junction temperature. As the flow rate increases further, the incremental temperature reduction diminishes, and the junction temperature asymptotically approaches a lower bound.

This nonlinear behavior indicates diminishing thermal benefit at higher coolant flow rates, beyond which additional flow yields limited improvement in junction temperature. The effect becomes more pronounced at higher power dissipation levels, where increasingly large increases in flow are required to achieve modest temperature reductions.

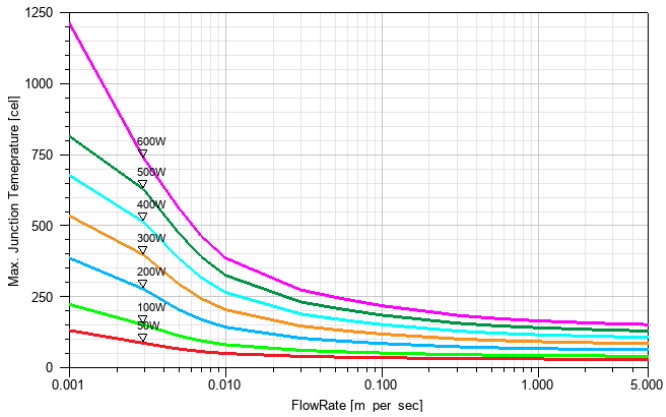

Fig. 7. Maximum junction temperature of a single SiC MOSFET on a three-MOSFET assembly as a function of coolant flow rate for selected power dissipation levels.

Using the complete parametric dataset, minimum coolant flow rates were extracted for specific junction temperature constraints. Figure 8 compares the minimum coolant flow rate required to maintain specified junction temperature limits for both a single MOSFET and the three-MOSFET configuration, assuming a constant conversion efficiency of 98%. The three-device curves reflect the increased total thermal load and corresponding scaling of cooling demand. The curves define a temperature-constrained operating envelope and highlight the strongly nonlinear increase in cooling demand as operation approaches stricter thermal limits.

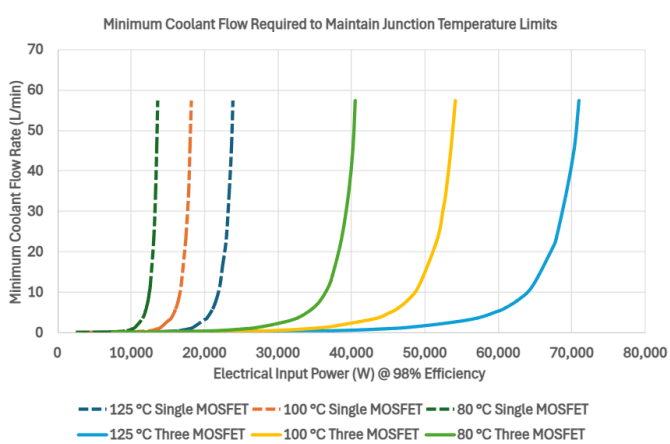

Fig. 8. Minimum coolant flow rate required to maintain the junction temperature below specified limits. Dashed lines correspond to a single SiC MOSFET, Solid lines correspond to a MOSFET in the three-device configuration.

To illustrate how the parametric simulation results can be used to extract practical operating points. Table II summarizes representative combinations of per-MOSFET heat dissipation, total electrical input power, target junction temperature, and the corresponding minimum coolant flow rate. Heat dissipation values are reported per MOSFET, while electrical input power represents the total input for the three-device configuration. These operating points were selected from the full parametric dataset to demonstrate the flexibility of the proposed framework across different thermal constraints. The table does not list all simulated cases but shows how the same results can be used to determine cooling requirements for different operating objectives.

TABLE II
REPRESENTATIVE OPERATING POINTS EXTRACTED FROM THE PARAMETRIC SIMULATIONS.

Heat Dissipated (W)	Electrical Input Power (W)	Target Junction Temperature (°C)	Required Flow Rate (L/min)
50	7,500	125	0.014
169.61	25,442	125	0.115
50	7,500	100	0.025
128.78	19,317	100	0.115
50	7,500	80	0.041
96.18	14,427	80	0.115
239.15	35,873	80	8.028

For stringent junction temperature limits at higher power levels, the required coolant flow increases sharply, as reflected by the highest-flow operating points in Table II. This behavior highlights practical cooling limitations when aggressive thermal constraints are imposed.

V. Predictive Maintenance Framework

The CFD-derived thermal maps developed in this work enable a model-based predictive thermal supervision framework for a three-MOSFET SiC power module configuration used in fast EV charging systems. By precomputing the relationship between per-module power dissipation, total electrical input, coolant flow rate, and junction temperature distribution, the thermal state of the multi-device assembly can be estimated without direct junction temperature sensing. Unlike a single-device system, the three-MOSFET configuration introduces thermal coupling effects due to shared cold-plate conduction paths and non-uniform coolant temperature rise along the flow direction.

During operation, real-time measurements of total electrical input power and coolant flow rate can be mapped onto the precomputed thermal characteristics. Using these maps, the junction temperature of each MOSFET can be inferred and continuously evaluated against predefined thermal limits. Figure 8 defines temperature-constrained operating envelopes for the complete three-device assembly, enabling system-level thermal supervision.

When operating conditions approach allowable thermal boundaries, preventive control actions—such as proportional power redistribution among modules, global power derating, coolant flow adjustment, or controlled shutdown—may be initiated. This approach enables early mitigation of excessive thermal stress while accounting for device-to-device thermal interaction.

Unlike conventional reactive protection mechanisms that rely solely on temperature threshold crossings, the proposed

framework enables proactive thermal management based on predicted multi-device junction temperature trends. Consequently, reliability, thermal balance, and operational availability can be enhanced in high-power fast-charging applications where aggregate heat density is significantly higher than in single-device systems.

VI. Conclusion

This paper presented a CFD-based thermal analysis of a three-SiC-MOSFET configuration mounted on a common cold plate for fast EV charging applications. Junction temperature behavior was quantified as a function of per-module power dissipation and coolant flow rate, while accounting for total system heat load and inter-device thermal coupling.

The results demonstrate a nonlinear increase in cooling demand as aggregate power dissipation increases. Thermal interaction between adjacent MOSFET modules further amplifies junction temperature rise, particularly at lower coolant flow rates, highlighting practical constraints on continuous high-power operation in multi-device assemblies.

By leveraging simulation-derived thermal maps for the complete three-device configuration, the proposed approach enables model-based prediction of individual junction temperatures under arbitrary operating conditions. This framework provides a practical foundation for system-level thermal supervision and preventive control strategies aimed at improving reliability, thermal uniformity, and operational robustness in high-power EV charging modules.

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